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Summary

My research focuses on evolutionary genomics, with particular interest in hybridization, lineage diversification, and trait evolution. I use integrative approaches that combine genomic data with comparative, morphological, and biogeographical analyses, grounded in open-source and reproducible workflows. My work aims to understand the evolutionary processes shaping adaptation, speciation, and biological diversity across both wild and agricultural systems.

Education

Doctor of Philosophy in Ecology, Evolution, and Behaviour 2020-2025

Concentration: Global Change Biology

Boise State University

Dissertation: *Understanding tropical plant evolution through hybridisation and adaptive introgression: a case study of Pandanaceae*

Doctoral Advisor: Professor Sven Buerki, PhD

Master of Science in Biology 2018-2020

Boise State University

Thesis: *A metanalysis of sequences of vascular plants in the world's biodiversity hotspots with a special section on Madagascar*

Major Advisor: Professor Sven Buerki, PhD

Honours Bachelor of Science in Marine Biology and Microbiology 2011-2016

Minors: French and Chemistry

Oregon State University

Honours Thesis: *Metagenomics and metatranscriptomics of the leaf-and root-associated microbiomes of Zostera marina and Zostera japonica*

Honours Advisor: Professor Byron C. Crump, PhD

Peer-Reviewed Publications

Wojahn JMA, Forest F, Maurin O, Smith J, Skinner P, Callmander MW, Buerki S. 2026. Adaptive introgression facilitates tropical plant evolution: *Pandanus* as a case study. *Molecular Phylogenetics and Evolution*. 108677. <https://doi.org/10.1016/j.ympev.2026.108677>

Wojahn JMA, Callmander MW, Buerki S. 2025. *Pandanus* Plastomes Decoded: When Climate Mirrors Morphology and Phylogenetic Relationships. *American Journal of Botany*. 111:e16461. <https://doi.org/10.1002/ajb2.16461>

Wojahn JMA, Callmander MW, Lowry PP, Buerki S. 2022. Update on the phylogenetic framework of Malagasy angiosperms. In S. M. Goodman (ed.), *The New Natural History of Madagascar*, pp. 464-470. Princeton University Press. <https://doi.org/10.2307/j.ctv2ks6tbb.58>

Buerki S, Philipson P, Lowry PP, **Wojahn JMA**, Andriambololonera S, Callmander MW. 2022. Sapindaceae In S. M. Goodman (ed.), *The New Natural History of Madagascar*, pp. 685-690. Princeton University Press. <https://doi.org/10.2307/j.ctv2ks6tbb.84>

Melton AE, Galla SJ, Dumanguit CD, **Wojahn JMA**, Novak S, Serpe M, Martinez P, Buerki S. 2022. Meta-Analysis Reveals Challenges and Gaps for Genome-to-Phenome Research Underpinning Plant Drought Response. *International Journal of Molecular Sciences*. 2022, 23(20), 12297 <https://doi.org/10.3390/ijms232012297>

Wojahn JMA, Galla SJ, Melton AE, Buerki S. 2021. G2PMineR: A Genome to Phenome Literature Review Approach. *Genes*. 12:293 <https://doi.org/10.3390/genes12020293>

Crump BC, **Wojahn JMA**, Tomas F, Mueller RS. 2018. Metatranscriptomics and Amplicon Sequencing Reveal Mutualisms in Seagrass Microbiomes. *Frontiers in Microbiology*. 9:388 <https://doi.org/10.3389/fmicb.2018.00388>

Manuscripts Submitted or in Preparation

Wojahn JMA, Smith J, Forest F, Maurin O, Nuraliev M, Poma M, Novak S, Besnard G, Callmander MW, Buerki S. 2026. Hybridisation and convergence drive repeated life-form innovation during environmental change. *In preparation for Proceedings of the Royal Society B*.

Research Experience

Independent Postdoctoral Researcher – Pandanaceae Project 2025–present

- Advanced phylogenomic investigations of Pandanaceae, spearheading new genomic analyses to investigate hybrid speciation and adaptive introgression dynamics in the tropics.
- Developed a novel nonparametric bootstrapping method and metric proof-of-concept for quantum phylogenetics, expanding the lab's methodological toolkit.
- Collaborated on a comparative plastome assembly study, designing and implementing Bayesian statistical analyses.
- Maintained active research momentum post-doctorate, independently managing project timelines, data workflows, and manuscript preparation for peer-reviewed publication.

Doctoral Researcher – Buerki Lab, Boise State University 2020–2025

- Led phylogenomic investigations of Pandanaceae: hybridisation, adaptive introgression, lineage diversification.
- Generated genome-scale datasets (target capture, plastomes) using reproducible bioinformatic workflows.
- Integrated molecular, morphological, and biogeographic data to address evolutionary and taxonomical questions.

EPSCoR Graduate Research Assistant – Boise State University 2020–2022

- Provided bioinformatics and programming support for NSF GEM3 project.
- Developed open-source bioinformatic tools and reproducible workflows for genome-to-phenome research.

Master's Researcher – Buerki Lab, Boise State University 2018–2020

- Examined reproducibility of collections and DNA sequences in biodiversity hotspots.

- Participated in field expedition to Madagascar.

Undergraduate Research – Crump Lab, Oregon State University 2012–2016

- Investigated *Zostera* microbiomes via metagenomics and metatranscriptomics.
- Collected microbial samples off Oregon, in the Columbia River, and in the Arctic Ocean.

Early Research Experience 2011–2012

- Fish Health Services, Lindsay Lab, Oregon State University – ELISA analyses on juvenile Pacific Salmon
- Hatfield Marine Science Center, Cheung Lab – fish egg enumeration

Consulting Experience

Owner/Operator of *Gnoll Pack Analytics, LLC* 2025-2026

- Operated an independent bioinformatics and ecological data consultancy delivering advanced quantitative solutions to research and environmental stakeholders in Boise, Idaho and remotely in Memphis, Tennessee.

Teaching Experience

- **Lab Instructor – Boise State University** 2018–2025
 - Introductory Cell and Molecular Biology
 - Developed G2PMineR lab module for Evolution & Ecology course (2022-2023)
- **Bioinformatics Instructor – EPSCoR** 2020
 - Mentored minority and first-generation students in R-based bioinformatics
- **Undergraduate French Teaching Assistant– Cheldelin Middle School** 2012-2014
 - Volunteered as a TA for the morning French class at CMS in Corvallis, Oregon.

Awards, Honours, and Professional Affiliations

- Award for Excellence in Teaching (Boise State University) 2024
- Graduate Showcase Dean’s Scholar (Boise State University) 2024
- Fellow, Linnean Society of London (FLS) Elected 2023
- Registered Professional Biologist (RPBio) British Columbia *Immediately Eligible*
- Articling Agrologist (AAg) British Columbia *Immediately Eligible*

Computational, Laboratory, and Other Experience

- **Scientific Software Development:** Design and maintenance of research-grade R packages and analytical pipelines.
- **Phylogenomics & Bioinformatics:** Target-capture data processing, model-based and coalescent phylogenetic inference, topology comparison, and model selection.
- **Genome Assembly & Comparative Genomics:** Organelle and draft genome assembly from short-read data, long-read genome assembly using PacBio HiFi data.
- **Data Science & Modeling:** Advanced statistical analysis in R; Unix-based workflows; C/C++, Python for machine learning and neural networks.
- **Trait Evolution & Biogeography:** Comparative methods, diversification analyses, molecular dating, and historical biogeographic modeling.

- **Molecular & Histological Methods:** DNA/RNA extraction, PCR, target-enrichment library preparation, gel-based assays, microtomy, and tissue staining
- **Plant Biology & Curation:** Morphometric analysis, *ex situ* cultivation, and herbarium specimen preparation.
- **Quantum & Advanced Computing:** Physical QPU execution (*Pegasus/Zephyr* topologies via D-Wave Leap), discrete optimization modeling, unconstrained quadratic programming, and high-performance recursive graph partitioning.
- **Languages:** English (native), French.

Equity, Diversity, and Inclusion

- **Departmental Representative & Board Member** 2024
 - Student Advisory Board for Inclusive Leadership
- **Professional Member** 2011-present
 - Out in Science, Technology, Engineering, and Mathematics (oSTEM)

Selected Software/Methods Developed

Wojahn, J.M.A., Ellestad, P.E. (2026) *Integrating Frequentist Bootstrapping with Quantum-Inspired Phylogenetic Reconstruction: A Proof-of-Concept*. github.com/wojahn/quantum-bootstrap-phylogenomics DOI: 10.5281/zenodo.20534276

Wojahn, J.M.A. (2025) *SoilSeeR*. github.com/wojahn/SoilSeeR. DOI: 10.5281/zenodo.20534270

Wojahn JMA, Galla SJ, Melton AE, Buerki S. (2022) *G2PMineR*. github.com/BuerkiLabTeam/G2PMineR DOI: 10.5281/zenodo.20583071

References

Professor Sven Buerki, PhD | Boise State University
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 (Responds in English or French)

Head Curator Martin Callmander, PhD | Conservatoire et Jardin botaniques de Genève
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 (Responds in English or French)

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